EMC TEST REPORT

Test report No:	**EMC-CE-3136**
Type of Equipment:	**ML-II distributed network**
Model Name:	**ML-II-DB96T, PCI-R1604 ML-II**
Applicant:	**AJINEXTEK CO., LTD.**
	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Manufacturer:	**AJINEXTEK CO., LTD.**
	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Test standards :	**EN 61000-6-4:2007**
	EN 61000-6-2:2005
Testing Laboratory:	**EMC Compliance Ltd.**
Test result :	**Complied**

This product complies with the requirements of the EMC Directive 2004/108/ EC.
The results in this report apply only to the sample tested.
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Date of receipt: 2012. 04. 10

Date of testing: 2012. 05. 01 ~ 05. 03

Issued date: 2012. 05. 08

Tested by: AHN, DO-WON

Approved by: YEOM, HAN-SEOK

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Contents

1. Applicant information 3

2. Laboratory information 4

3. Test system configuration 5

3.1 Operation environment 5

3.2 Measurement Uncertainty 6

4. Description of E.U.T. 7

4.1 General information 7

4.2 Product description 9

4.3 Auxiliary equipments 9

4.4 Test configuration 10

4.5 Operating conditions 11

5. Summary of test results 12

5.1 Modification to the E.U.T. 12

5.2 Summary of EMI emission test results 12

5.3 Summary of immunity test results 12

5.4 Performance criteria 13

6. Test results 14

6.1 Conducted Emission 14

6.2 Radiated Emission 19

6.3 Electrostatic Discharge 23

6.4 Radio Frequency Electromagnetic Fields 29

6.5 Electric Fast Transient/BURST 32

6.6 Surge 35

6.7 Conducted Immunity 38

6.8 Magnetic field immunity 41

6.9 Dips and Interruptions 43

7. E.U.T. photographs 45

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1. Applicant information

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Fax:	+82-53-593-3703
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Contact name:	**Lee Seung-Kyu**

Manufacturer:	AJINEXTEK CO., LTD.
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E-mail:	skleeok7@ajinextek.com
Contact name:	**Lee Seung-Kyu**

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2. Laboratory information

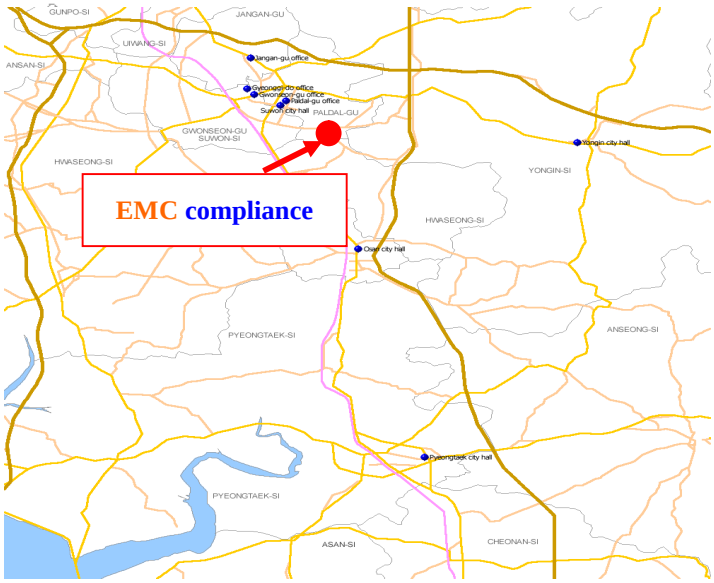
Address

EMC compliance Ltd.
480-5 Sin-dong, Yeongtong-gu, Suwon-city, Gyeonggi-do, 443-390, Korea
Telephone Number: 82 31 336 9919
Facsimile Number: 82 31 336 4767

FCC CAB.: KR0040
VCCI Registration No. : C-1713, R-2710
Industry Canada Registration No.: 8035
KOLAS NO.: 231

SITE MAP





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3. Test system configuration

3.1 Operation environment

		Temperature	Humidity	Pressure
Chamber(10 m)	:	24 °C	32 % R.H.	-
Shielded room(CE)	:	25 °C	36 % R.H.	-
Shielded room(ESD)	:	25 °C	37 % R.H.	100.2 kPa

Test site

These testing items were performed following locations;

Test item	Test site
Conducted Emission	Shielded Room
Discontinuous interference	Shielded Room
Radiated Emission	10 m Chamber
Harmonics current	Immunity area
Voltage fluctuations and flickers	Immunity area
Electrostatic discharge	Shielded Room
Radiated RF immunity	Fully anechoic chamber (3 m)
Electric Fast Transient/BURST	Shielded Room
Surge	Shielded Room
Conducted RF immunity	Shielded Room
Magnetic field immunity	Shielded Room
Voltage dip/interruption	Shielded Room

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3.2 Measurement Uncertainty

All measurements involve certain levels of uncertainties, especially in field of EMC.
The factors contributing to uncertainties are test receiver, cable loss, antenna factor calibration, Antenna directivity, antenna factor variation with height, antenna phase center variation, antenna frequency interpolation, measurement distance variation, site imperfection, mismatch, and system repeatability. Based on CISPR 16-4-2, the measurement uncertainty level with a 95 % confidence level was applied.

Conducted emission measurement :(k = 2, 95 %)		
Shielded Room (CE#1)	9 ㎑ ~ 150 ㎑: ± 3.77 [dB] 150 ㎑ ~ 30 ㎒: ± 3.35 [dB]	
Shielded Room (CE#2)	9 ㎑ ~ 150 ㎑: ± 3.95 [dB] 150 ㎑ ~ 30 ㎒: ± 3.55 [dB]	
Shielded Room (CE#3)	9 ㎑ ~ 150 ㎑: ± 3.88 [dB] 150 ㎑ ~ 30 ㎒: ± 3.50 [dB]	
Radiated Emission measurement :(k = 2, 95 %)		
10 m Chamber (#F4)	30 ㎒ ~ 300 ㎒	3 m: + 4.29 [dB], - 4.31 [dB] 10 m: + 4.28 [dB], - 4.30 [dB]
	300 ㎒ ~ 1 000 ㎒	3 m: + 4.57 [dB], - 4.57 [dB] 10 m: + 4.42 [dB], - 4.44 [dB]
	1 ㎓ ~ 6 ㎓	3 m: + 5.94 [dB], - 5.95 [dB]
10 m Chamber (#F2)	30 ㎒ ~ 300 ㎒	3 m: + 4.28 [dB], - 4.30 [dB] 10 m: + 4.27 [dB], - 4.28 [dB]
	300 ㎒ ~ 1 000 ㎒	3 m: + 4.56 [dB], - 4.57 [dB] 10 m: + 4.41 [dB], - 4.43 [dB]
	1 ㎓ ~ 6 ㎓	3 m: + 5.94 [dB], - 5.95 [dB]
Radio Frequency Electromagnetic Fields :(k = 2, 95 %)		
± 1.82 [dB]		
Disturbance power Electromagnetic Fields: (k = 2, 95 %)		
± 3.73 [dB]		

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4. Description of E.U.T.

4.1 General information

1. ML-II-DB96T Specifications

Module Function	MECHATROLINK-II based slave 96CH Digital Input/Output function module
Node ID Setting	LSB hexadecimal number rotary switch (0~F), MSB switch (0~1)
Mounting Method	Rear Panel - M3 Bolt 4 point mounting
Power supply / Current Consumption	DC +24V / Max. 500mA (TB1 Connection to Outside Module, I/O power except)
LED display	SlaveCPU : Power(yellow), Live(green), Error1(red), Error2(red) DB16T : Power(yellow), Link(green), Error(red), Com(green) INOUT1 ~ INOUT96(yellow)
Digital Input/Output	96 Input/Output channel, 24V DC Level IN1~IN96 : Photo-coupler isolation (MIN. 2mA / channel) OUT1~OUT96 : Photo-coupler + Drive IC (MAX. 50mA / channel) PCOM, NCOM : Digital I/O Positive and Negative Common Connector : HIROSE (PCN10A-48P-2.54DS)
Operating Temperature / Humidity	0 ~ 55 ℃ / 0 ~ 90% (No condensation)
Storage Temperature	-20 ~ 70 ℃
Weight	940g
Exterior dimension of product	151 (H) mm X 121 (D) mm X 100 (W) mm

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2. PCI-R1604 ML-II Specifications

Communication type	MECHATROLINK-II (YASKAWA distributed network specification)
Communication port	2 port (Up to 15 slaves can be installed on each port)
Physical Layer	RS485
Transmission speed	10 Mbps
Transmission cycle time	0.5mSec, 0.25mSec selectable
Transmission bytes	17 byte / 32 byte
Control axis	16 axes BASE, Max. 30 axes (Option Contact Board)
Max. Number of nodes	30 nodes
Max. Install board of system	8 boards
Motion control chipset	CAMC-QI 4EA (Expend CAMC-QI 4EA)
Velocity of motion command	~10.0Mpps (Standard of Clock 39.3216MHz)
LED display	All port : Comm (green), Error (red)
Max. Number of configuration slaves	30 slaves
Range of motion command	32-bit Range Counter (command/actual equal)
Bus Interface	PCI 2.2 Memory Access, 32bit
Network Connection	USB type A, dedicated STP cable (C.I. : 130Ω) Min. 0.5M (between nodes), Max. 50M (total)
Power supply / Current consumption	DC 5V / Max. 1.5A
Operating temperature / Humidity	0 ~ 60 ℃ / 0 ~ 80% (No condensation)
Storage temperature	-20 ~ 70 ℃
Weight	123g (Expend 30 axes – 172g)
Exterior dimension of product	174.63 (W) X 106 (L) mm

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4.2 Product description

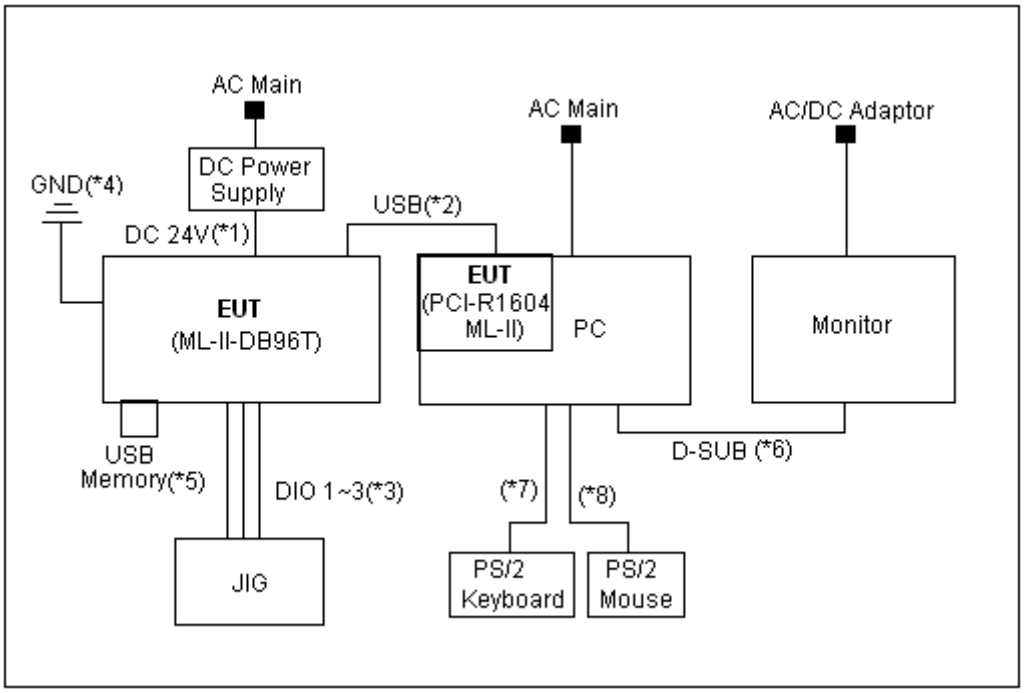
Type of product	ML-II distributed network
Model name (Basic)	ML-II-DB96T, PCI-R1604 ML-II
Model name (Variant)	N/A
Difference	-
Trade name	-
Serial no	Engineering Sample
Testing voltage	ML-II-DB96T: DC 24 Ⅴ PCI-R1604 ML-II: 230 Ⅴ, 50 Hz (Power supplied from PC)
Product rating	ML-II-DB96T: DC 24 Ⅴ PCI-R1604 ML-II: DC 5 Ⅴ
Internal clock frequency	39.3216 ㎒
Note	This test was done with ML-II-DB96T, PCI-R1604 ML-II.

4.3 Auxiliary equipments

Type	Model / Part #	Serial number	Manufacturer
PC	PCI-610BP-26-EA	-	-
Monitor	M15S-AN	N093N4KN800734	SAMSUNG
DC Power Supply	6032A	US38322201	HP
PS/2 Keyboard	SEM-DT35	3V001028	SAMSUNG
PS/2 Mouse	SMOP5000WX	4080016300	-
JIG	-	-	-
USB Memory	W6022	-	-

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4.4 Test configuration



Note	Start		End		Cable	
*	Name	I/O port	Name	I/O port	Length (m)	Spec.
1	**EUT** (ML-II-DB96T)	Power	DC Power Supply	Power	1.8	Non-Shield
2		USB	EUT (PCI-R1604 ML-II)	USB	2.5	Shield
3		DIO 1~3	JIG	DIO 1~3	3.0	Shield
4		GND	GND	GND	2.0	Non-Shield
5		USB	USB Memory	USB	Direct	-
6	PC	D-SUB	Monitor	D-SUB	1.5	Shield
7		PS/2	PS/2 Keyboard	PS/2	1.6	Non-Shield
8		PS/2	PS/2 Mouse	PS/2	1.7	Non-Shield

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4.5 Operating conditions

The EUT was configured as normal intended use.

Test mode	Normal operation
1	Operating mode. (Program: EZ ConFig)

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5. Summary of test results

5.1 Modification to the E.U.T.

None

5.2 Summary of EMI emission test results

Applied	Test items	Test method	Result
☒	Conducted Emission	EN 61000-6-4:2007	Complied
☐	Discontinuous interference	EN 61000-6-4:2007	N/A
☒	Radiated Emission	EN 61000-6-4:2007	Complied
☐	Harmonics current	EN 61000-3-2:2006+A2:2009	N/A
☐	Voltage fluctuations and flickers	EN 61000-3-3:2008	N/A

5.3 Summary of immunity test results

Applied	Test items	Test method	Result
☒	Electrostatic discharge	EN 61000-4-2:2009	Complied
☒	Radiated RF immunity	EN 61000-4-3:2006+A1:2008	Complied
☒	Electric Fast Transient/BURST	EN 61000-4-4:2004+A1:2010	Complied
☒	Surge	EN 61000-4-5:2006	Complied
☒	Conducted RF immunity	EN 61000-4-6:2009	Complied
☒	Magnetic field immunity	EN 61000-4-8:2010	Complied
☒	Voltage dip/interruption	EN 61000-4-11:2004	Complied

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5.4 Performance criteria

The Variety and the diversity of the apparatus within the scope of this standard makes it difficult to define precise criteria for the evaluation of the immunity test results.
If, as a result of the application of the tests defined in this standard, the apparatus becomes dangerous or unsafe, the apparatus shall be deemed to have failed the test.
A functional description and a definition of performance criteria, during or as a consequence of the EMC testing, shall be provided by the manufacturer and noted in the report, based on the following criteria.

Performance criterion A: The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by an permissible loss of performance.
If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

Performance criterion B: The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation, and from what the user may reasonably expect from the apparatus if used as intended.

Performance criterion C: Temporary loss of function is allowed, provided the function is self-recoverable of can be restored by the operation of the controls.

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6. Test results

6.1 Conducted Emission

Test specification	EN 61000-6-4:2007		
Testing voltage	230 V, 50 Hz		
Test facility	Shielded room (CE#1)		
Date	2012. 05. 01		
Temperature (°C)	25 °C	Humidity (% R.H.)	36 % R.H.
Remarks	Complied		

6.1.1 Limits of conducted emission measurement

☒ AC main

Frequency [MHz]	Limit values (dB(μV))	
	Quasi-peak	Average
0.15 ~ 0.5	79	66
0.5 ~ 30	73	60

☐ Telecommunication

Frequency [MHz]	Voltage Limits (dB(μV))		Current Limits (dB(μA))	
	Quasi-Peak	Average	Quasi-Peak	Average
0.15 ~ 0.5	97 to 87	84 to 74	53 to 43	40 to 30
0.5 ~ 30	87	74	43	30

* The limits decrease linearly with the logarithm of the frequency in the range 0.15 MHz to 0.5 MHz
* The current and voltage disturbance limits are derived for use with an impedance stabilization Network (ISN) which presents a common mode (asymmetric mode) impedance of 150 Ω to the telecommunication port under test (conversion factor is 20 $\log_{10}$ 150/I = 44 dB).

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6.1.2 Measurement procedure

The measurements were performed in a shielded room. EUT was setup as shown in photograph and placed on a non-metallic table height of 0.8 m above the reference ground plane. The rear of table was located 0.4 m to the vertical conducted plane. EUT was power through the LISN, which was bonded to the ground plane. The LISN power was filtered. Each EUT power lead, except ground (safety) lead was individually connected through a LISN to input power source. EUT signal cables that hung closer than 0.4 m to the Horizontal metal ground 0.3 m ~ 0.4 m long. The power cord was bundles in the center. All peripheral equipment was powered from a sub LISN. The LISN and ISN were positioned 0.8 m from the EUT. Peak and Average detection were used in preliminary testing and Quasi-peak and Average detections were used at final measurement.

6.1.3 Used equipments

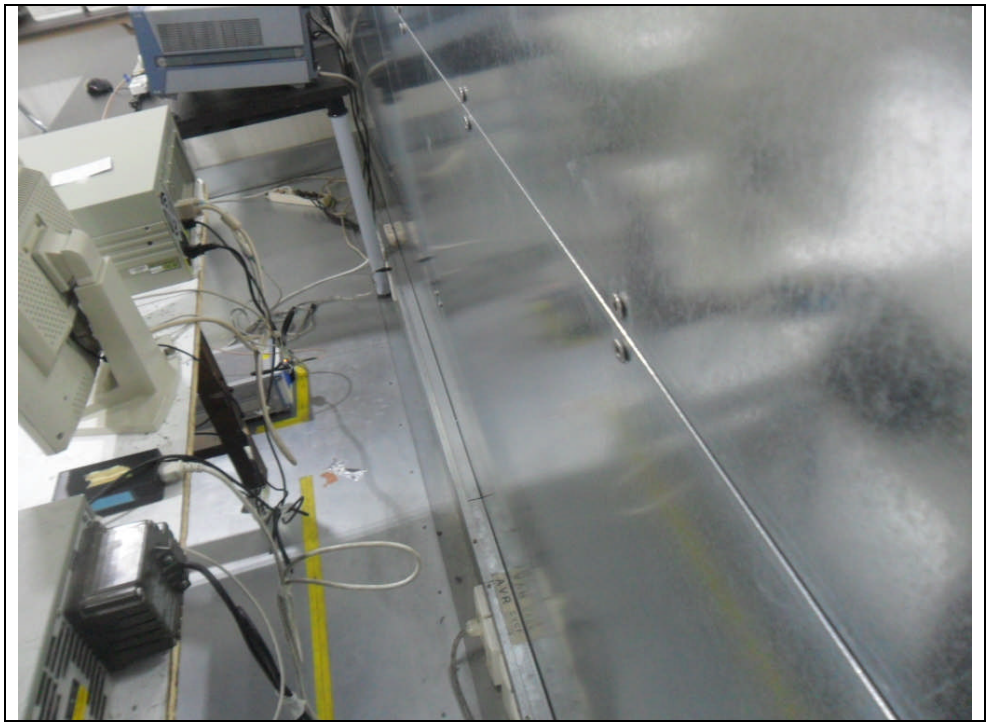
Equipment	Model	Serial No.	Makers	Next Cal. Date	Used
Test Receiver	ESHS30	828765/009	R&S	2012.10.28	☒
LISN	ENV216	101358	R&S	2012.10.26	☒
LISN	L2-16A	0000J10705	PMM	-	☒
ISN	T800	24314	TESEQ	2012.12.12	☐
8-WIRE ISN	NTFM 8158 CAT5	CAT5-8158-0028	SCHWARZBECK	2013.04.06	☐
8-WIRE ISN	NTFM 8158 CAT3	CAT3-8158-0020	SCHWARZBECK	2013.04.06	☐
ISN	ENY81	101545	R&S	2012.09.01	☐

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6.1.4 Photographs of test setup

* AC Main





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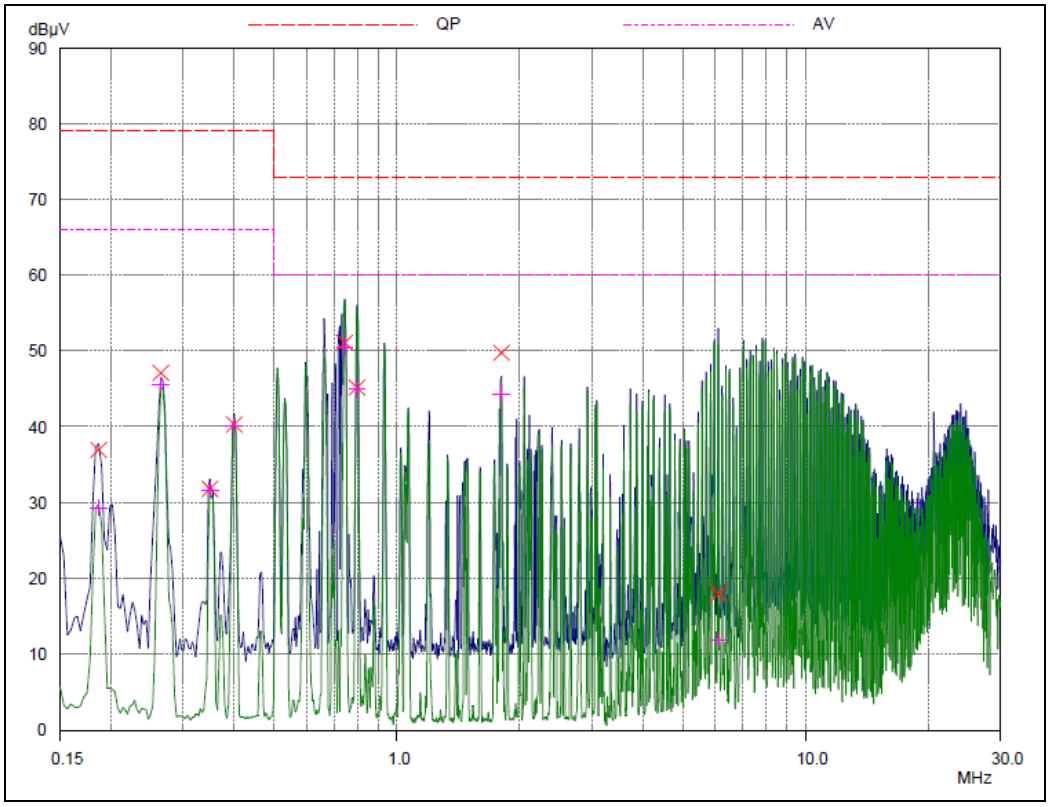
6.1.5 Conducted emission measurement result

* AC Main

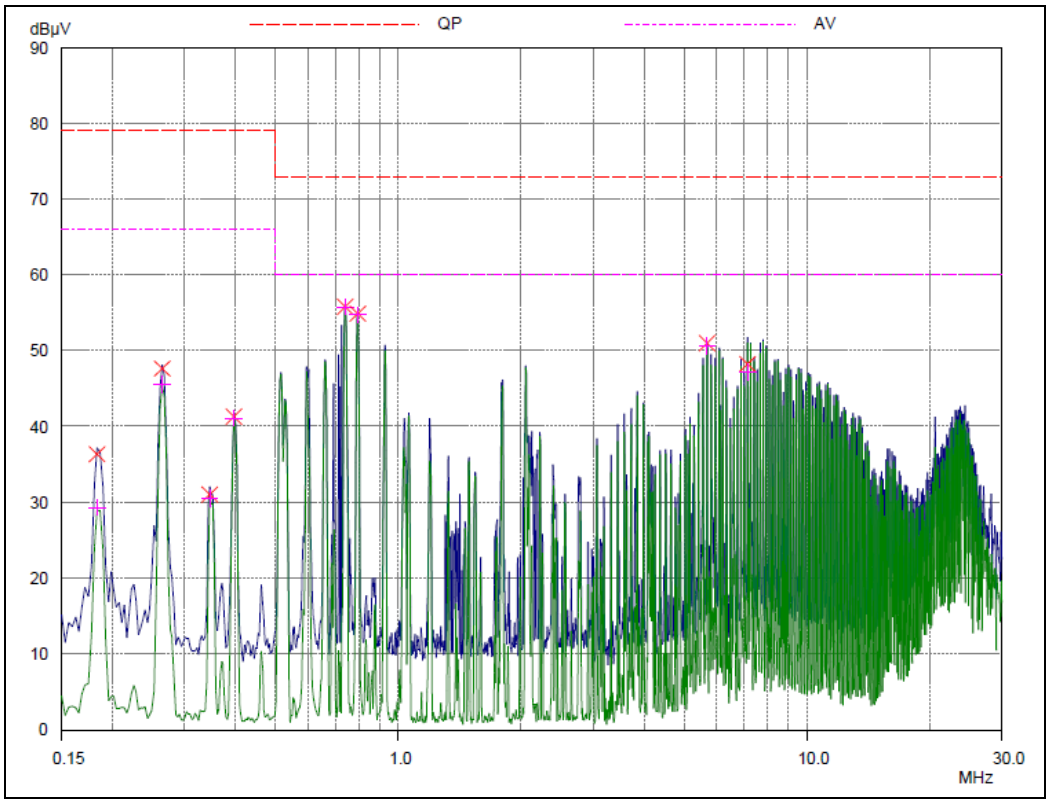
Frequency	Correction Factor		Line	Quasi-peak				Average			
				Limit	Reading	Result	Margin	Limit	Reading	Result	Margin
[㎒]	LISN	Cable		[dB(㎶)]	[dB(㎶)]	[dB(㎶)]	[dB]	[dB(㎶)]	[dB(㎶)]	[dB(㎶)]	[dB]
0.183	9.90	0.01	N	79.00	26.46	36.37	42.63	66.00	19.41	29.32	36.68
0.186	9.90	0.01	H		27.05	36.96	42.04		19.44	29.35	36.65
0.264	9.90	0.01	N		37.74	47.65	31.35		35.70	45.61	20.39
0.348	9.93	0.01	H		21.87	31.81	47.19		21.76	31.70	34.30
0.396	9.94	0.01	N		31.36	41.31	37.69		31.19	41.14	24.86
0.399	9.96	0.01	H		30.34	40.31	38.69		30.05	40.02	25.98
0.741	9.88	0.02	N	73.00	45.90	55.80	17.20	60.00	45.92	55.82	4.18
0.744	9.88	0.02	H		41.23	51.13	21.87		40.59	50.49	9.51
0.795	9.84	0.02	N		45.05	54.91	18.09		44.92	54.78	5.22
1.800	9.66	0.02	H		40.08	49.76	23.24		34.62	44.30	15.70
5.690	9.62	0.05	N		41.37	51.04	21.96		40.97	50.64	9.36
7.150	9.63	0.06	N		38.62	48.31	24.69		37.51	47.20	12.80

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[Hot-Line]



[Neutral-Line]



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6.2 Radiated Emission

Test specification	EN 61000-6-4:2007		
Testing voltage	DC 24 Ⅴ / 230 Ⅴ, 50 ㎐		
Test facility	10 m Chamber (#F2)		
Test distance	10 m		
Date	2012. 05. 01		
Temperature (°C)	24 °C	Humidity (% R.H.)	32 % R.H.
Remarks	Complied		

6.2.1 Limits of radiated emission measurement

Frequency [㎒]	(dB(㎶/m)) @10 m
30 ~ 230	40
230 ~ 1 000	47

6.2.2 Measurement procedure

The test was done at a 10 m chamber with a quasi-peak detector. EUT was placed on a non-metallic table height of 0.8 m above the reference ground plane. Cables were folded back and forth forming a bundle 0.3 m to 0.4 m long and were hanged at a 0.4 m height to the ground plane.
Cables connected to EUT were fixed to cause maximum emission.
Test was made with the antenna positioned in both the horizontal and vertical planes of polarization.
The measurement antenna was varied in height above the conducting ground plane to obtain the maximum signal strength.

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6.2.3 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. Date	Used
Test Receiver	ESCI	100710	R&S	2012.11.28	☒
Bi-Log Antenna	VULB 9168	9168-440	SCHWARZBECK	2013.10.04	☒
Amplifier	310N	293004	SONOMA INSTRUMENT	2012.11.28	☒
3 dB Attenuator	8491A	27444	HP	2012.11.28	☒
Antenna Mast	MA4000-EP	303	Innco Systems	-	☒
Turn Table	DT2000S-1t	079	Innco Systems	-	☒
Horn ANT	3115	00086706	ETS	2013.11.21	☐
Amplifier	8449B	3008A01802	AGILENT	2012.05.11	☐
Spectrum Analyzer	FSP7	100289	R&S	2012.12.19	☐

6.2.4 Sample calculation

The field strength is calculated adding the antenna Factor, cable loss and, Antenna pad adding, subtracting the amplifier gain from the measured reading.

The sample calculation is as follow:

Result = M.R + C.F(A.F + C.L +3 dB Att – A.G)

M.R = Meter Reading

C.F = Correction Factor

A.F = Antenna Factor

C.L = Cable Loss

A.G= Amplifier Gain

3 dB Att = 3 dB Attenuator

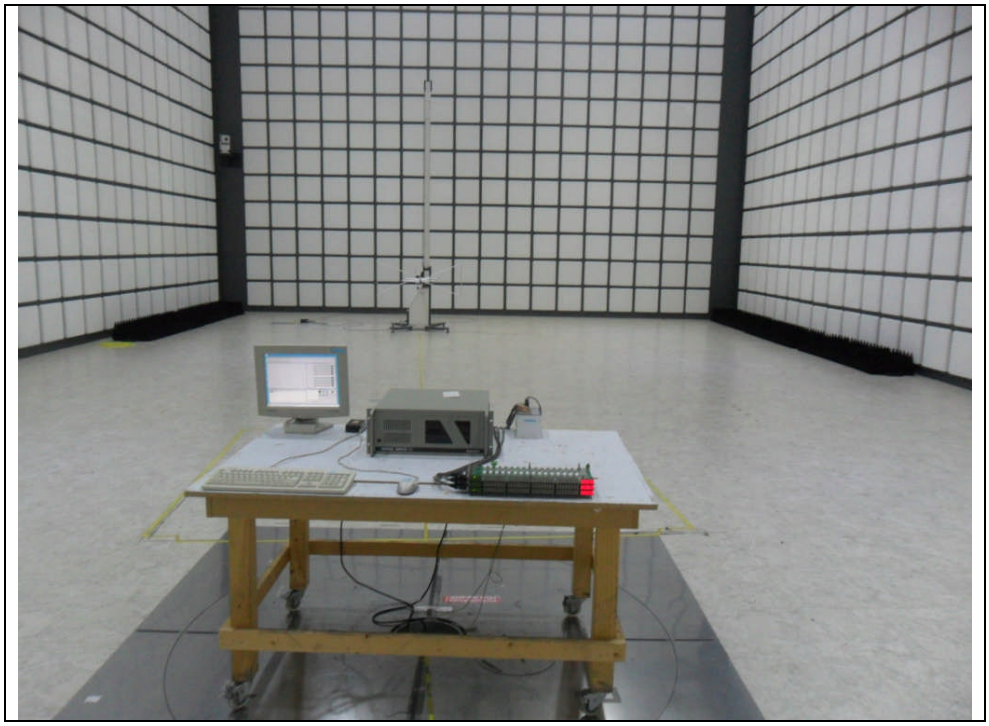
If M.R is 30 dB, A.F 12 dB, C.L 5 dB, 3 dB, A.G 35 dB

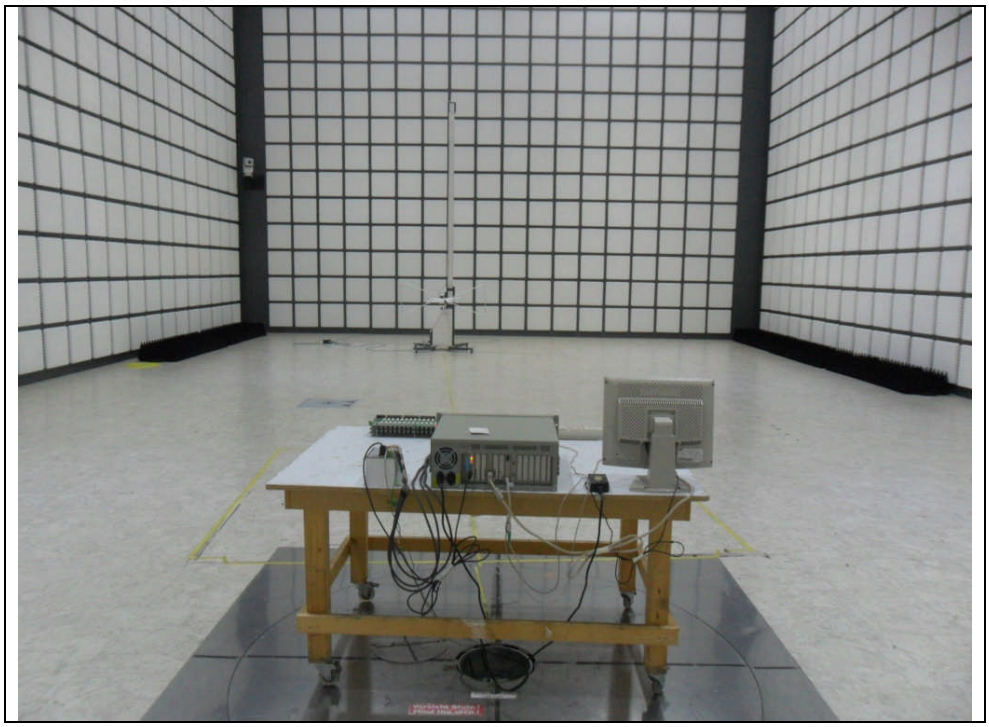
The result is30 + 12 + 5 + 3 – 35 = 15 dB(μV/m)

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6.2.5 Photographs of test setup

* DC 24 V / 230 V, 50 Hz



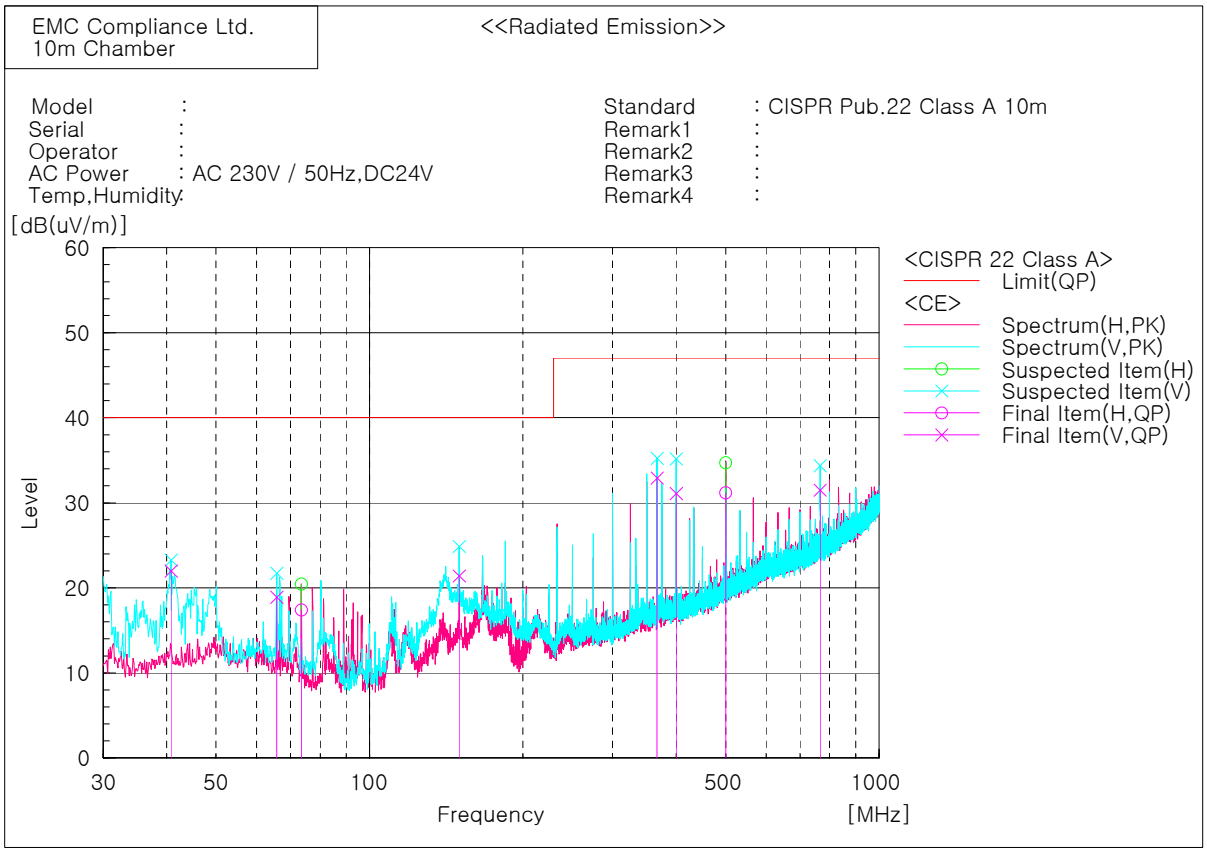


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6.2.6 Radiated emission measurement result

*** Graph and Data**

* DC 24 V / 230 V, 50 Hz



Final Result

No.	Frequency [MHz]	(P)	Reading QP [dB(uV)]	c.f [dB(1/m)]	Result QP [dB(uV/m)]	Limit QP [dB(uV/m)]	Margin QP [dB]	Height [cm]	Angle [deg]
1	40.791	V	36.5	-14.5	22.0	40.0	18.0	198.0	255.0
2	65.769	V	33.9	-15.0	18.9	40.0	21.1	298.0	3.5
3	73.523	H	34.0	-16.6	17.4	40.0	22.6	400.0	252.4
4	149.931	V	33.9	-12.5	21.4	40.0	18.6	101.0	76.4
5	366.723	V	41.6	-8.7	32.9	47.0	14.1	101.0	19.4
6	399.954	V	39.0	-7.9	31.1	47.0	15.9	101.0	130.8
7	499.976	H	36.5	-5.3	31.2	47.0	15.8	202.0	285.9
8	766.742	V	31.8	-0.3	31.5	47.0	15.5	400.0	353.6

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6.3 Electrostatic Discharge

Test specification	EN 61000-4-2:2009, Criteria : B				
Test level	☒ Contact: ± 2 ㎸, ± 4 ㎸ ☒ Air: ± 2 ㎸, ± 4 ㎸, ± 8 ㎸ ☒ HCP: ± 2 ㎸, ± 4 ㎸ ☒ VCP: ± 2 ㎸, ± 4 ㎸				
Discharge impedance	330 Ω / 150 ㎊				
Number of discharge (Each polarity)	☒ Contact: 25 ☒ Air: 10 ☒ HCP / VCP: 25				
Interval between discharges	1 s				
Testing voltage	DC 24 Ⅴ / 230 Ⅴ, 50 ㎐				
Test facility	Shielded room				
Date	2012. 05. 02				
Temperature(°C)	25 °C	Humidity (% R.H.)	37 % R.H.	Pressure (㎪)	100.2 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.3.1 Measurement procedure

A ground reference plane was located on the floor, and connected to earth via a low Impedance connection. The return cable of the ESD generator was connected to the reference plane.
In case of floor standing equipment, EUT was placed on the reference plane on 0.1 ㎡ of insulating Support. In case of table top equipment, EUT was placed on a wooden table 0.8 ㎡ above the reference grounded floor. A horizontal coupling plane (HCP) was placed on the table, and Connected to the reference plane via a 470 ㏀ resistor located in each end (0.5 ㎜ insulating support between EUT and HCP). In both cases a vertical coupling plane(VCP) OF 0.5 X 0.5 ㎡ was located 0.1 ㎡ from the EUT's sides. The VCP was connected to the reference plane in the same matter as the HCP.

6.3.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. Date	Used
ESD Tester	NSG 437	182	TESEQ	2012.05.23	☒
HCP	-	-	-	-	☒
VCP	-	-	-	-	☒

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6.3.3 Photographs of test setup

* DC 24 V



* 230 V, 50 Hz



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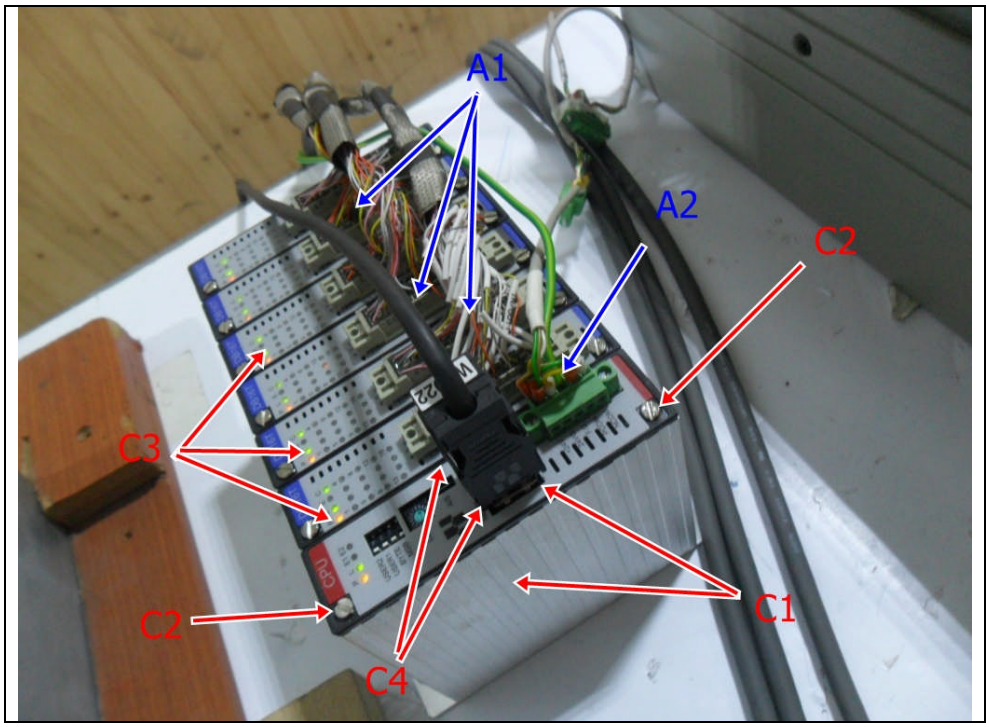
6.3.4 Measurement result

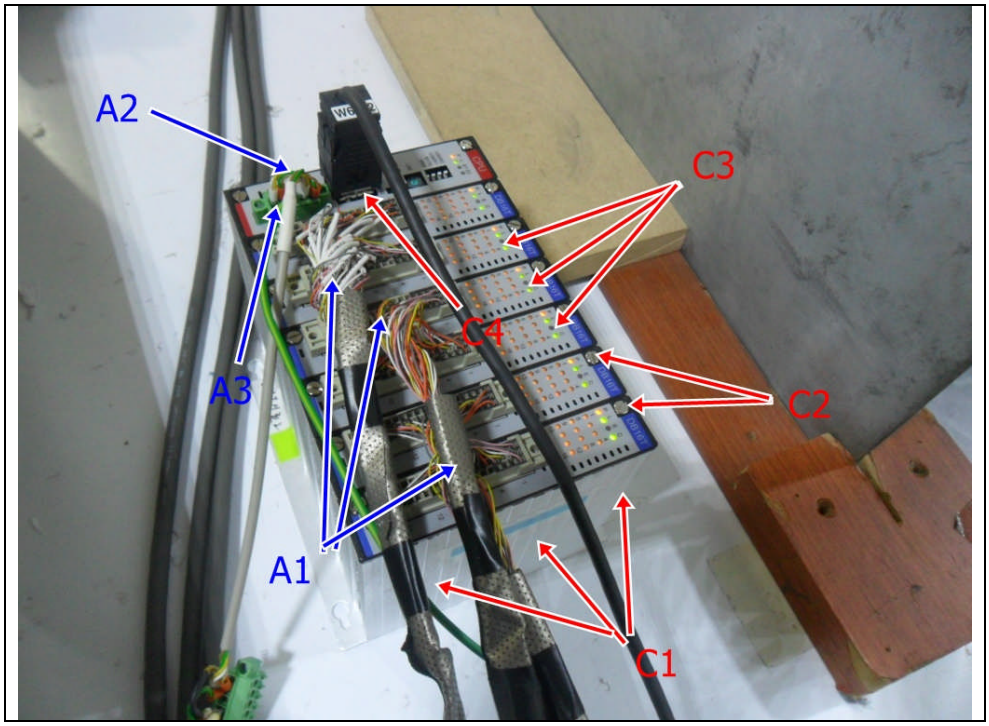
Electrostatic Discharge (Test Point)

Air discharge →

Contact discharge →

* DC 24 V





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HCP/VCP discharge

Location(EUT)	Applied level (±)	Result (Criterion)
HCP (All 4 sides)	± 2 kV, ± 4 kV	A
VCP (All 4 sides)	± 2 kV, ± 4 kV	A

Contact discharge

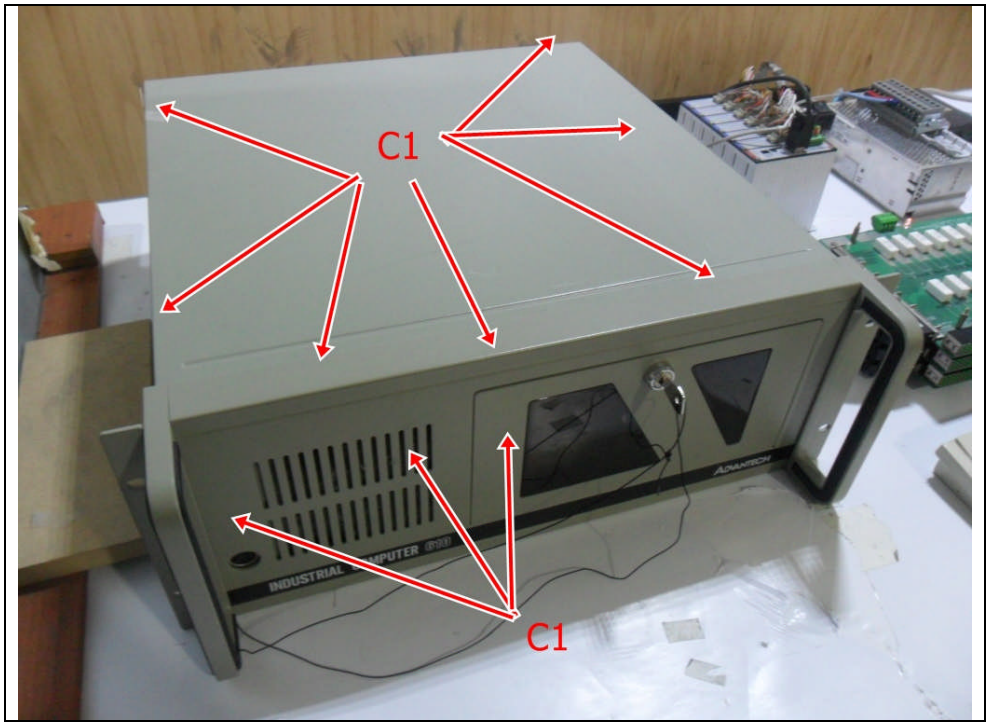
Location(EUT)		Applied level (±)	Result (Criterion)
C1	Enclosure(Case)	± 2 kV, ± 4 kV	A
C2	Screw	± 2 kV, ± 4 kV	A
C3	LED Around	± 2 kV, ± 4 kV	A
C4	USB Port	± 2 kV, ± 4 kV	A

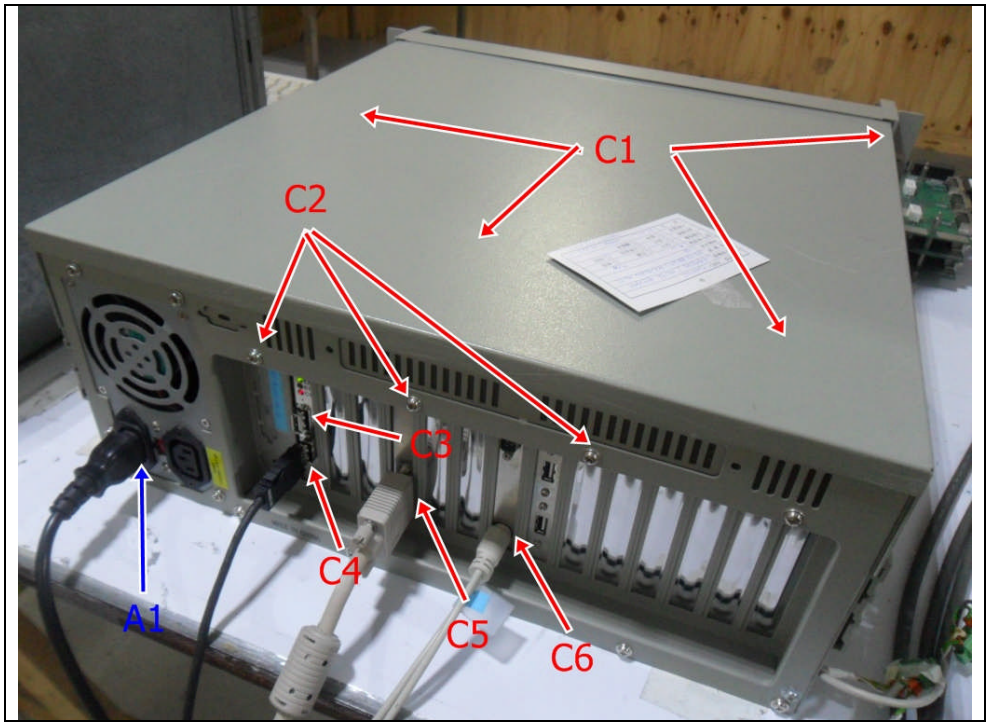
Air discharge

Location(EUT)		Applied level (±)	Result (Criterion)
A1	DIO Port	± 2 kV, ± 4 kV, ± 8 kV	A
A2	Power Port	± 2 kV, ± 4 kV, ± 8 kV	A
A3	GND	± 2 kV, ± 4 kV, ± 8 kV	A

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* 230 V, 50 Hz





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HCP/VCP discharge

Location(EUT)	Applied level (±)	Result (Criterion)
HCP (All 4 sides)	± 2 kV, ± 4 kV	A
VCP (All 4 sides)	± 2 kV, ± 4 kV	A

Contact discharge

Location(EUT)		Applied level (±)	Result (Criterion)
C1	Enclosure(Case)	± 2 kV, ± 4 kV	A
C2	Screw	± 2 kV, ± 4 kV	A
C3	Bracket	± 2 kV, ± 4 kV	A
C4	USB Port	± 2 kV, ± 4 kV	A
C5	D-SUB Port	± 2 kV, ± 4 kV	A
C6	PS/2 Port	± 2 kV, ± 4 kV	A

Air discharge

Location(EUT)		Applied level (±)	Result (Criterion)
-	-	-	-

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6.4 Radio Frequency Electromagnetic Fields

Test specification	EN 61000-4-3:2006+A1:2008, Criteria : A				
Tested frequency	80 ㎒ ~ 1 ㎓, 1.4 ㎓ ~ 2 ㎓, 2 ㎓ ~ 2.7 ㎓				
Test level & Modulation	80 ㎒ ~ 1 ㎓: 10 V/m, 80 % Amplitude Modulation (1 ㎑) 1.4 ㎓ ~ 2 ㎓: 3 V/m, 80 % Amplitude Modulation (1 ㎑) 2 ㎓ ~ 2.7 ㎓: 1 V/m, 80 % Amplitude Modulation (1 ㎑)				
Frequency Step	log 1 % step				
Dwell time	3 s				
Distance	3 ㎜ from EUT to tip of antenna				
Testing Voltage	DC 24 Ⅴ / 230 Ⅴ, 50 ㎐				
Test facility	Fully anechoic chamber (3 ㎜)				
Date	2012. 05. 02				
Temperature(°C)	25 °C	Humidity (% R.H.)	39 % R.H.	Pressure (㎪)	100.2 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.4.1 Measurement procedure

The test was performed at 3 ㎜ full anechoic chamber.

For floor standing equipment, the EUT was standing on the floor.

For tabletop equipment, the EUT was located on a wooden table 0.8 ㎜ above the floor.

The EUT was tested all sides, horizontal and vertical polarization.

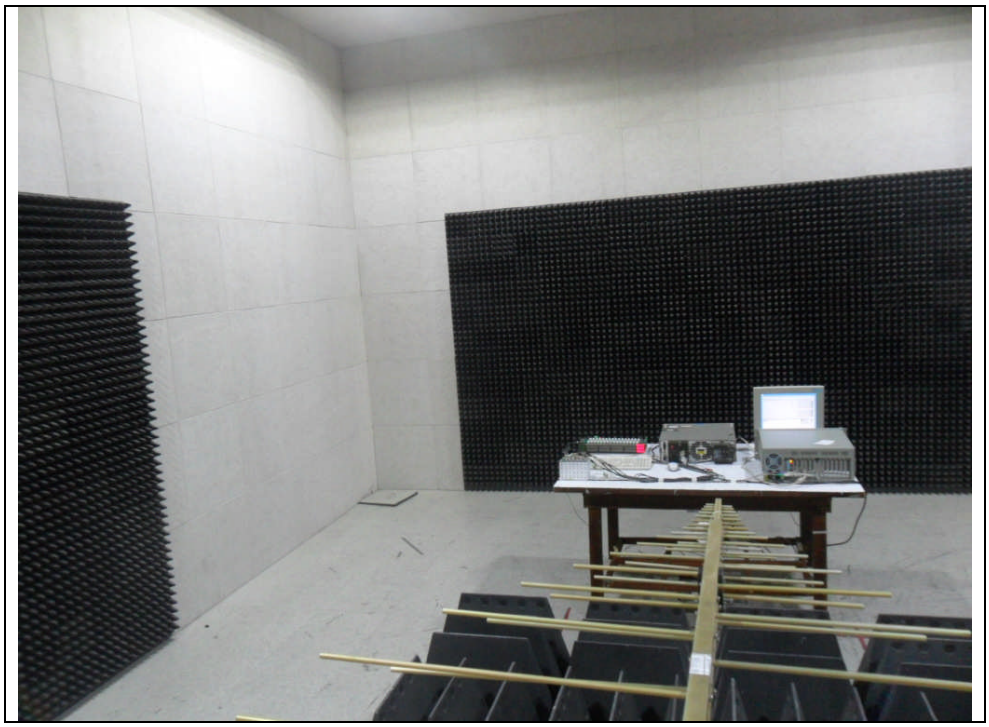
This test report shall not be reproduced, except in full, without the written approval.

6.4.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Power meter	PM2002	302852	AR	2013.04.06	☒
Power sensor	PH2000	303224	AR	2013.04.06	☒
Power sensor	PH2000	311217	AR	2013.04.06	☒
Directional coupler	DC6180	303976	AR	2013.04.06	☒
Directional coupler	DC7144M1	320279	AR	2013.02.28	☒
Signal generator	E4421B	GB40052295	AGILENT	2012.10.12	☒
Amplifier	BBA100	100996-1	R&S	2013.02.07	☒
Amplifier	60S1G3M2	320444	AR	2013.04.06	☒
Broadband Ant.	LPDA-0803	130269	ETS	-	☒
Fiber Optic Modem	HI-4413P	-	ETS-LINDGREM	-	☒
Antenna master	-	-	ETS	-	☒

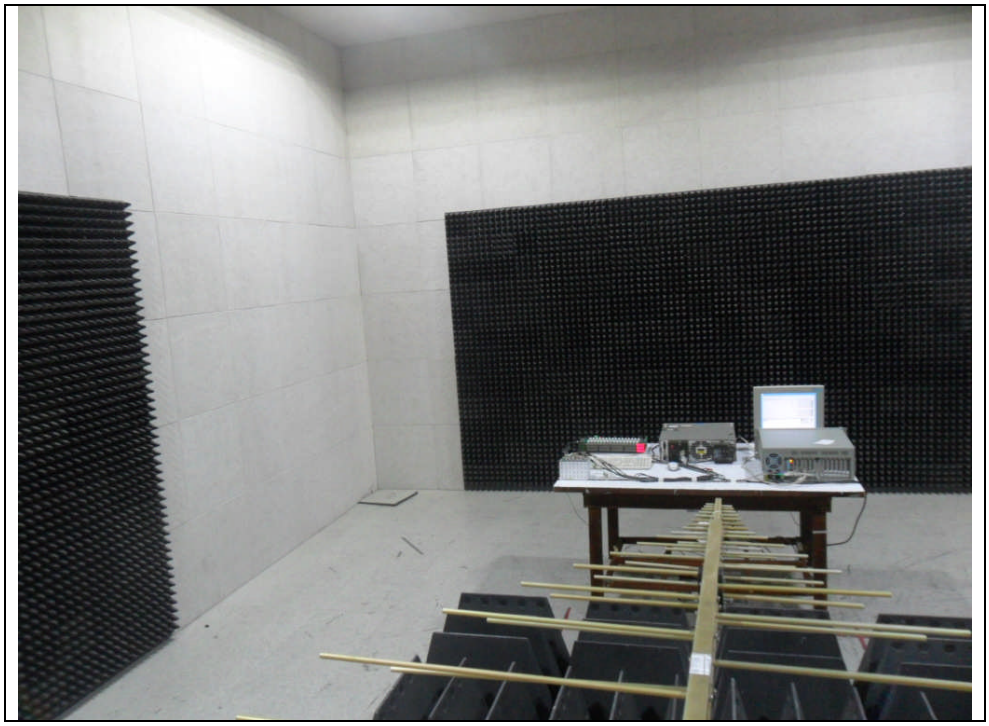
6.4.3 Photographs of test setup

* DC 24 V



This test report shall not be reproduced, except in full, without the written approval.

* 230 V, 50 Hz



6.4.4 Measurement result

* DC 24 V / 230 V, 50 Hz

Location(EUT)	Antenna polarization	Result (Criterion)
Front side	Horizontal	A
	Vertical	A
Rear side	Horizontal	A
	Vertical	A
Left side	Horizontal	A
	Vertical	A
Right side	Horizontal	A
	Vertical	A

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6.5 Electric Fast Transient/BURST

Test specification	EN 61000-4-4:2004+A1:2010, Criteria : B				
Coupling	☒ DC 24 V / AC main ☒ Signal: Clamp				
Test level	☒ DC / AC Power: ± 2 kV Peak ☒ Signal: ± 1 kV Peak				
Repetition frequency	5 kHz, Tr/Th = 5 / 50 ns				
Coupling time	60 s				
Testing Voltage	DC 24 V / 230 V, 50 Hz				
Test facility	Shielded room				
Date	2012. 05. 03				
Temperature(°C)	24 °C	Humidity (% R.H.)	38 % R.H.	Pressure (kPa)	100.1 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.5.1 Measurement procedure

A ground reference plane was located on the floor.
EFT generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.1 m wooden table.
For tabletop equipment, EUT was placed on a 0.1 m above the ground reference plane.
Test generator and coupling/decoupling network was placed on, and bounded to, the ground reference plane. When using the coupling clamp, the minimum distance between the coupling plates and all other conductive surfaces, except the ground reference plane beneath the coupling clamp, Shall be 0.5 m.

6.5.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	2013.02.08	☒
Capacitive Coupling Clamp	-	-	EM TEST	-	☒

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6.5.3 Photographs of test setup

* DC 24 V



* 230 V, 50 Hz



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* Signal (DC 24 V)



6.5.4 Measurement result

* DC Line

EFT coupling point	(+)	(-)	Result (Criterion)
DC 24 V	+ 2 kV	- 2 kV	A

* AC main

EFT coupling point	(+)	(-)	Result (Criterion)
L+N+PE	+ 2 kV	- 2 kV	A

* Signal (DC 24 V)

EFT coupling point	(+)	(-)	Result (Criterion)
DIO	+ 1 kV	- 1 kV	A

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6.6 Surge

Test specification	EN 61000-4-5:2006, Criteria : B				
Coupling	☒ DC 24 V / AC main: Direct ☐ Signal: Direct / CDN				
Test level	☒ AC main: ☒Differential mode: ± 0.5 kV, ± 1 kV ☒Common mode: ± 0.5 kV, ± 1 kV, ± 2 kV ☒ DC 24 V : ± 0.5 kV ☐ Signal: ± 0.5 kV, ± 1 kV				
Coupling Impedance	☒ Differential mode: 18 μF ☒ Common mode: 10 Ω + 9 μF ☐ 40 Ω ☒ Direct				
Surge pulse shape	Tr/Th = 1.2 / 50 μs				
Angles	0 °, 90 °, 180 °, 270 °				
Number of surge	5				
Coupling time	1 min				
Testing Voltage	DC 24 V / 230 V, 50 Hz				
Test facility	Shielded room				
Date	2012. 05. 03				
Temperature(°C)	24 °C	Humidity (% R.H.)	38 % R.H.	Pressure (kPa)	100.1 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.6.1 Measurement procedure

A ground reference plane was located on the floor. SURGE generator was connected to reference ground plane via low impedance connection. For floor standing equipment & table top equipment, EUT was placed on a wooden table.

6.6.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	2013.02.08	☒
CDN	CNV 508 S1	0302-01	EM TEST	-	☐

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6.6.3 Photographs of test setup

* DC 24 V



* 230 V, 50 Hz



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6.6.4 Measurement result

* DC Line

Coupling point	(+)	(-)	Result (Criterion)
DC 24 V	+ 0.5 kV	- 0.5 kV	A

* AC main

Coupling point	(+)	(-)	Result (Criterion)
L+N	+ 0.5 kV, + 1 kV	- 0.5 kV, - 1 kV	A
L+PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A
N+PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A

* Signal

Coupling point	(+)	(-)	Result (Criterion)
-	-	-	-

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6.7 Conducted Immunity

Test specification	EN 61000-4-6:2009, Criteria : A				
Tested frequency	0.15 ㎒ ~ 80 ㎒				
Test level & Modulation	10 Ⅴ, 80 % Amplitude Modulation (1 ㎑)				
Frequency Step	log 1 % step				
Coupling method	☒ DC 24 Ⅴ / AC main: CDN ☒ Signal: Clamp				
Testing Voltage	DC 24 Ⅴ / 230 Ⅴ, 50 ㎐				
Test facility	Shielded room				
Date	2012. 05. 02				
Temperature(°C)	26 °C	Humidity (% R.H)	37 % R.H	Pressure(㎪)	100.2 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.7.1 Measurement procedure

A ground reference plane was located on the floor.
The test was performed on a ground reference plane on a 0.1 m wooden table.
This test were performed using CDN for mains, clamp for signal and injection probe.
The frequency range was swept from 0.15 ㎒ to 80 ㎒. This frequency range was
Modulated with 1 ㎑ sine wave at 80 %.
The signal generators provided the modulated frequency at a 1 % step size.
The power and all network cable, I/O cables longer than 3 m length were tested.

6.7.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
CS generator	CWS 500 C S1	V0635101750	EM TEST	2012.10.13	☒
CDN	CDN M2/M3	0906-12	EM TEST	2012.10.12	☒
CDN	CDN M3-32A	0506-29	EM TEST	2013.02.07	☒
Attenuator	73-6-34	MU918	MCE/WEINSCHEL	2012.10.13	☒
EM Clamp	KEMZ 801	17643	Schaffner	2012.04.25	☒
Current probe	MD720	W1345167/M6/ 0068	Schaffner	-	☐

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6.7.3 Photographs of test setup

* DC 24 V



* 230 V, 50 Hz



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* Signal (DC 24 Ⅴ)



6.7.4 Measurement result

* DC Line

Coupling point	Coupling method	Result (Criterion)
DC 24 Ⅴ	CDN(M2)	A

* AC main

Coupling point	Coupling method	Result (Criterion)
Power	CDN(M3)	A

* Signal (DC 24 Ⅴ)

Coupling point	Coupling method	Result (Criterion)
DIO	Clamp	A

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6.8 Magnetic field immunity

Test specification	EN 61000-4-8:2010, Criteria : A				
Magnetic field strength	30 A/m (rms)				
Coupling time	60 s				
Frequency	50 ㎐				
Polarization	X, Y, Z				
Testing voltage	230 Ⅴ, 50 ㎐				
Test facility	Shielded room				
Test Date	2012. 05. 03				
Temperature (°C)	24 °C	Humidity (% R.H.)	38 % R.H.	Pressure (㎪)	100.1 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.8.1 Measurement procedure

The test was performed on a ground reference plane (GRP) on a 0.8 m wooden table.
The EUT was isolated 10 cm isolating support. The ground plane was connected to floor reference ground plane via low impedance connection. The test generator was placed 3 m distance from the induction coil. The generator was connected reference ground plane. Preliminary verification of equipment performances was carried out prior to applying the test magnetic field. The field was applied to the EUT horizontal, vertical polarization.

6.8.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Ultra compact simulator	UCS 500 M6	V0545100858	EM TEST	2013.02.08	☒
Magnetic coil	MS 100	0701-03	EM TEST	-	☒
Current transformer	MC 2630	113-97	EM TEST	-	☒

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6.8.3 Photographs of test setup

* 230 V, 50 Hz



6.8.4 Measurement result

* 230 V, 50 Hz

Positions	Coil Polarity	Test level	Frequency	Result (Criterion)
Enclosure	X	30 A/m	50 Hz	A
	Y			A
	Z			A

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6.9 Dips and Interruptions

Test specification	EN 61000-4-11:2004 , Criterion: B or C				
Number of dips	3 T				
Duration	60 s				
Phase	Zero crossing (0 °)				
Testing Voltage	230 Ⅴ, 50 ㎐				
Test facility	Shielded room				
Test Date	2012. 05. 03				
Temperature (°C)	24 °C	Humidity (% R.H)	38 % R.H	Pressure (㎪)	100.1 ㎪
Remarks	Complied				

6.9.1 Measurement procedure

The dips/interruption test is only applicable to AC mains.

The dips/interruptions were applied at zero crossing.

6.9.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Generator	UCS 500 M6	V0545100858	EM TEST	2013.02.08	☒

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6.9.3 Photographs of test setup

* 230 V, 50 Hz



6.9.4 Measurement result

*230 V, 50 Hz

Test Level (%UT)	Dip/Int. (%UT)	Duration /Period	Angle (°)	Count number	Result
0 %	100 %	1 Period	0	3T	A
40 %	60 %	10 Period	0	3T	C
70 %	30 %	25 Period	0	3T	C
0 %	100 %	250 Period	0	3T	C

Comment:

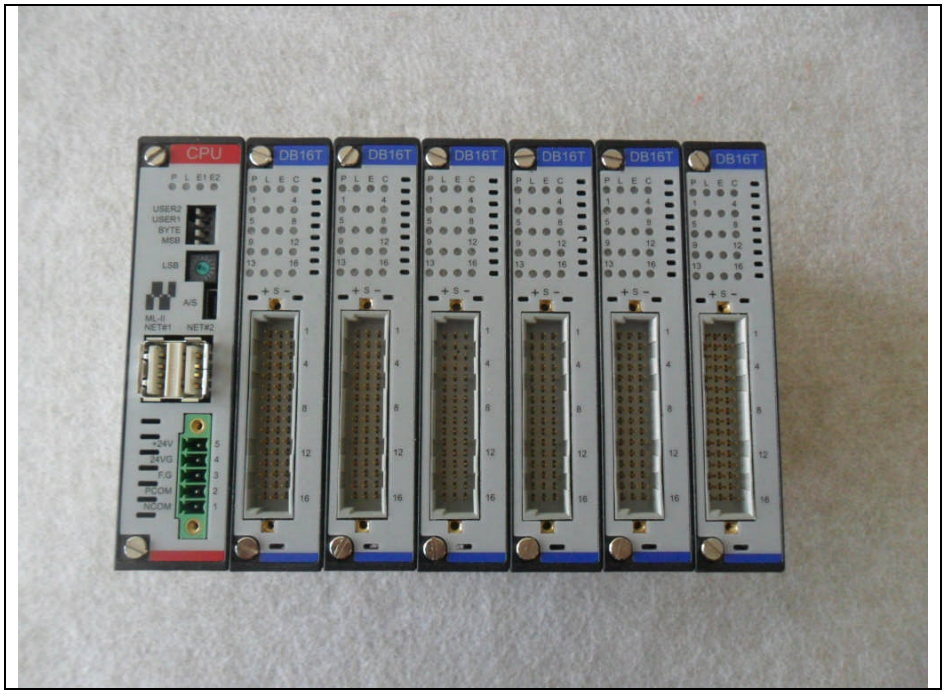
- A: There was no change of operation status during above testing. (1 Period)
- C: The power of PC& EUT was turned off as soon as the test started.
  After the test, EUT was getting back to normal operation (10 /25 / 250 Period)

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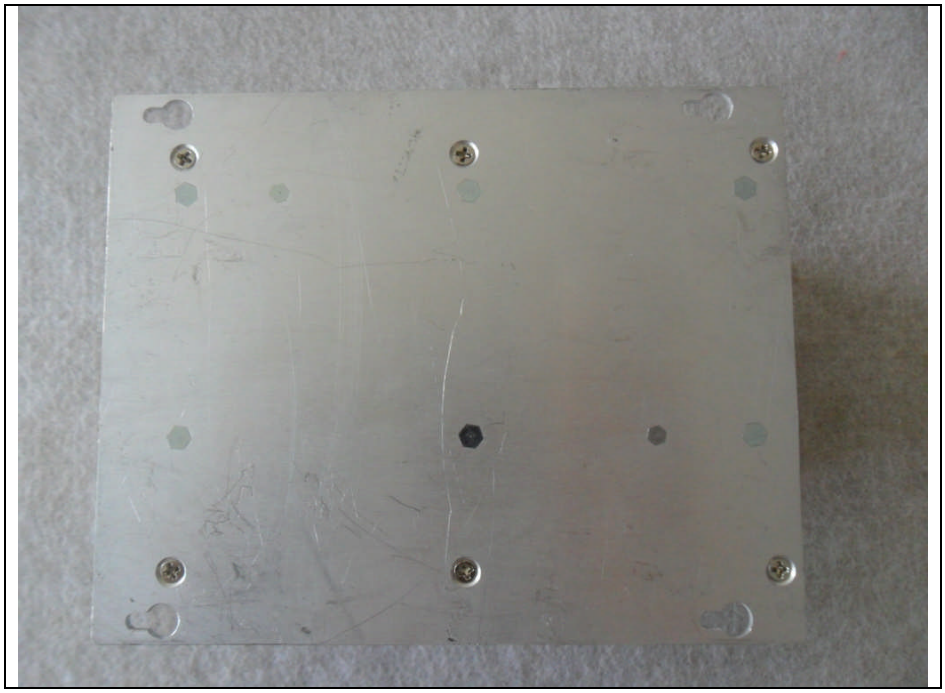
7. E.U.T. photographs

* ML-II-DB96T

Front View

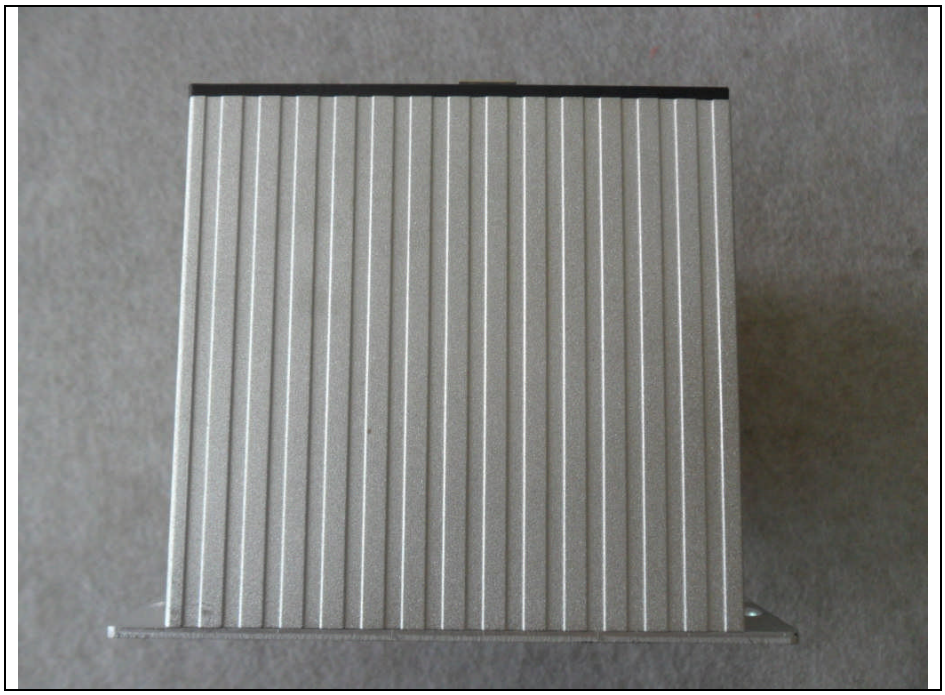


Rear View

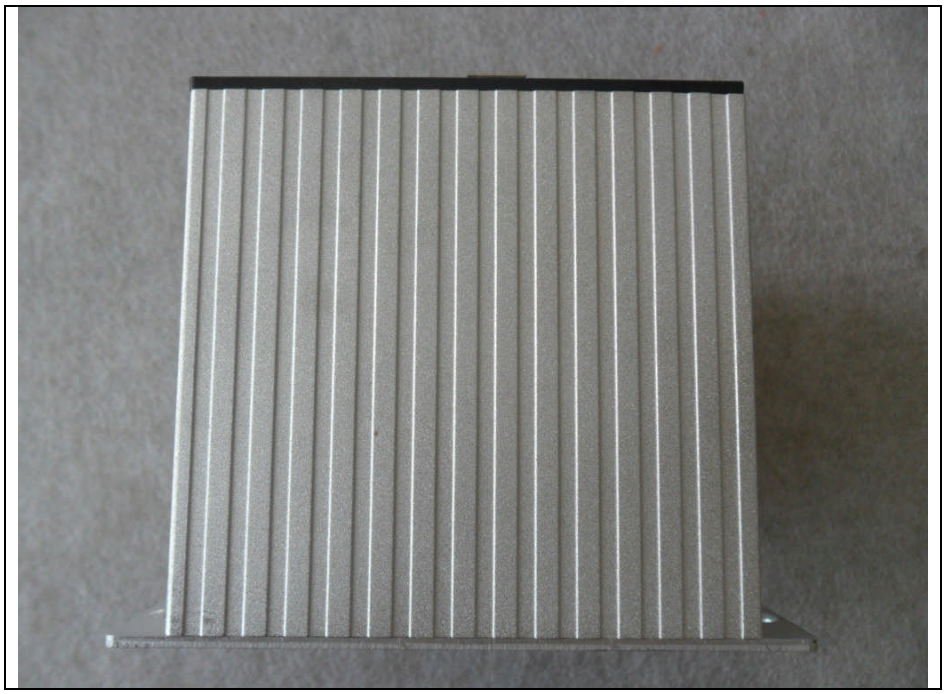


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Left View

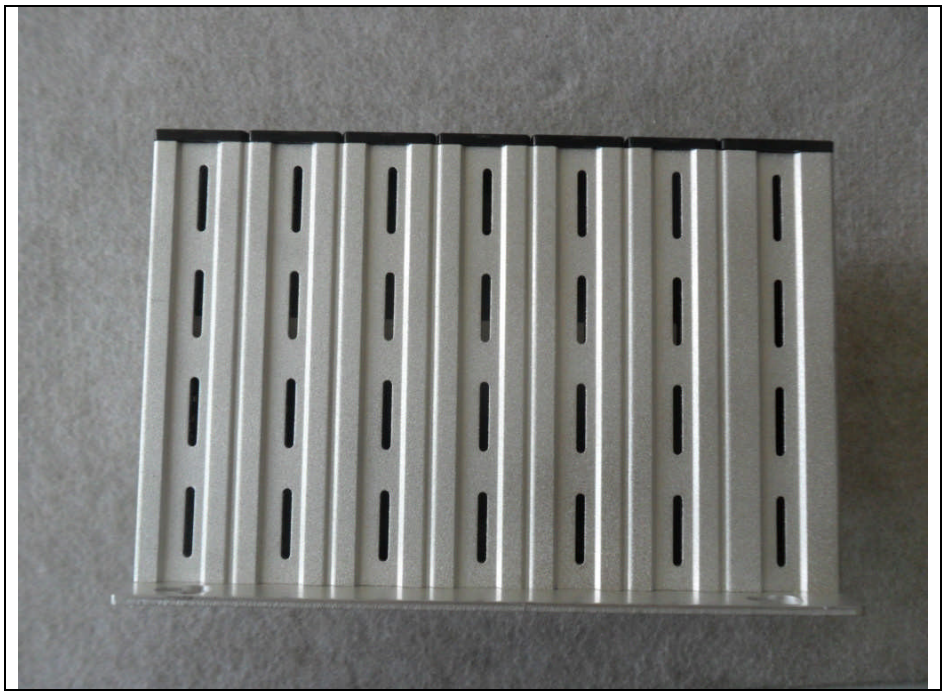


Right View

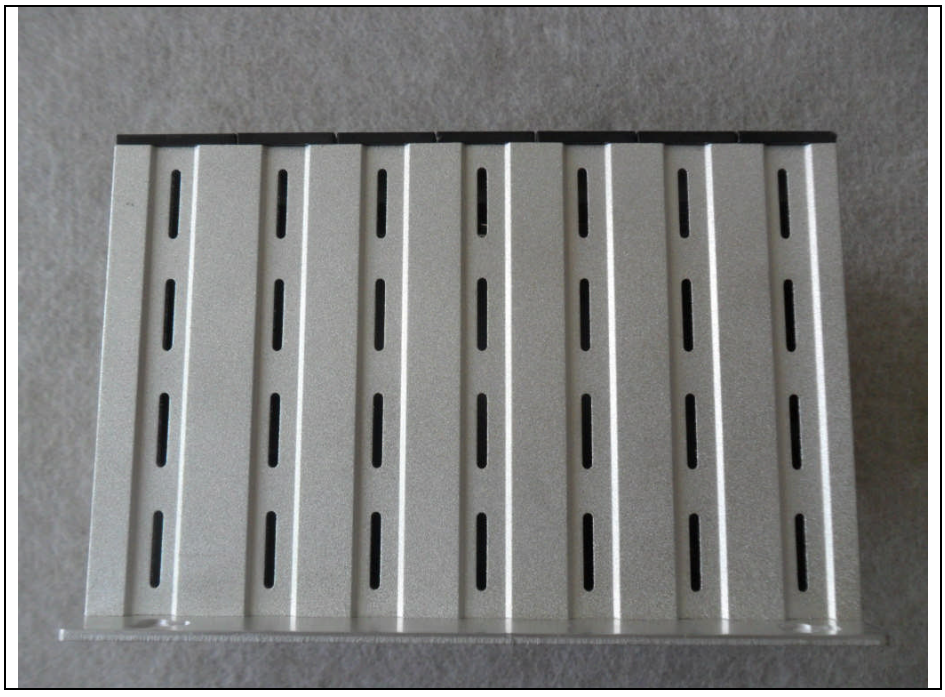


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Top View

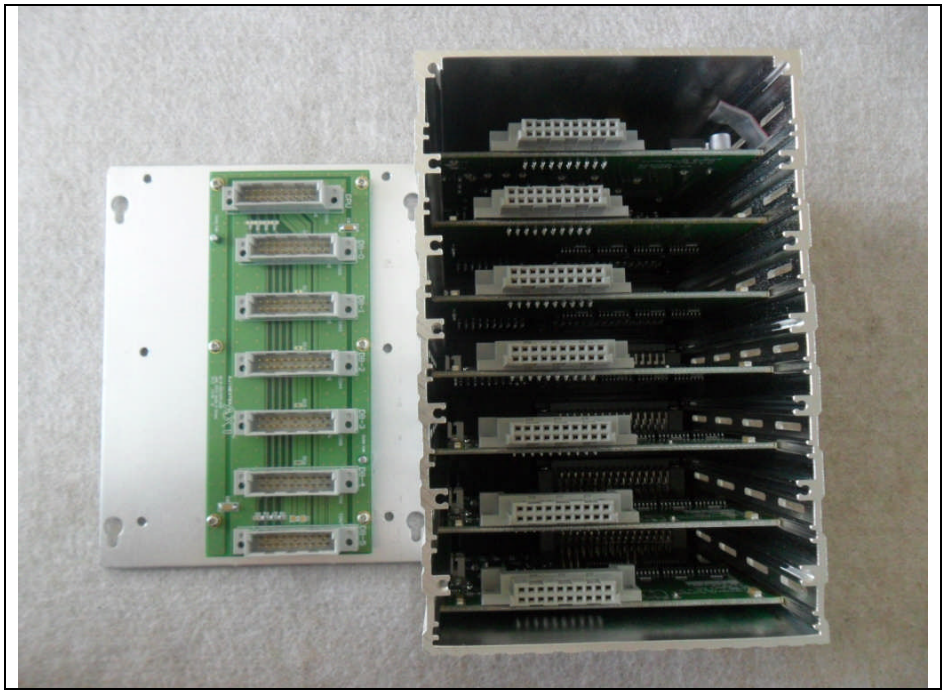


Bottom View



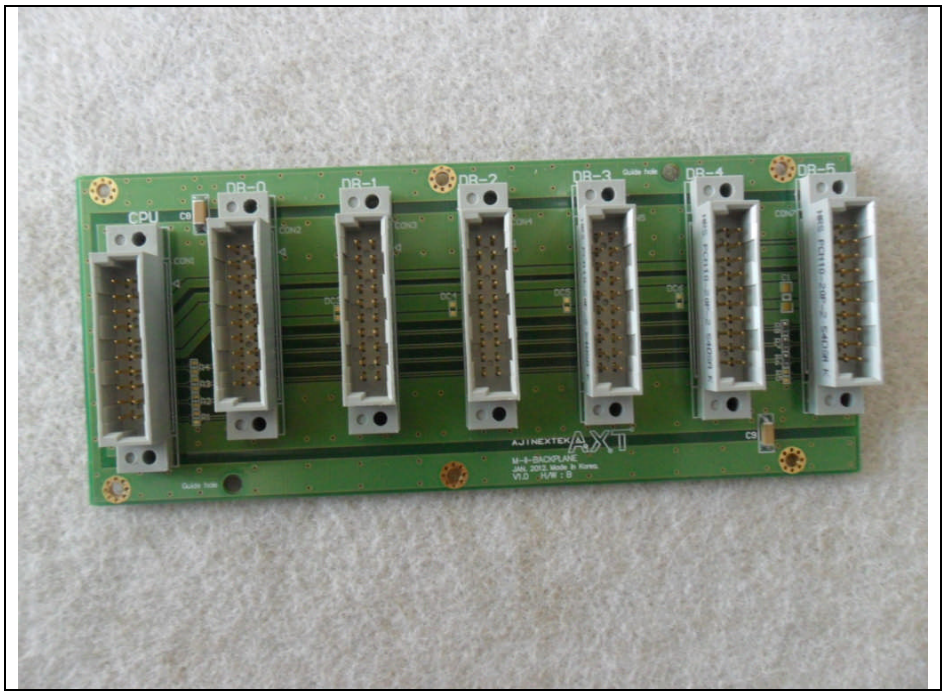
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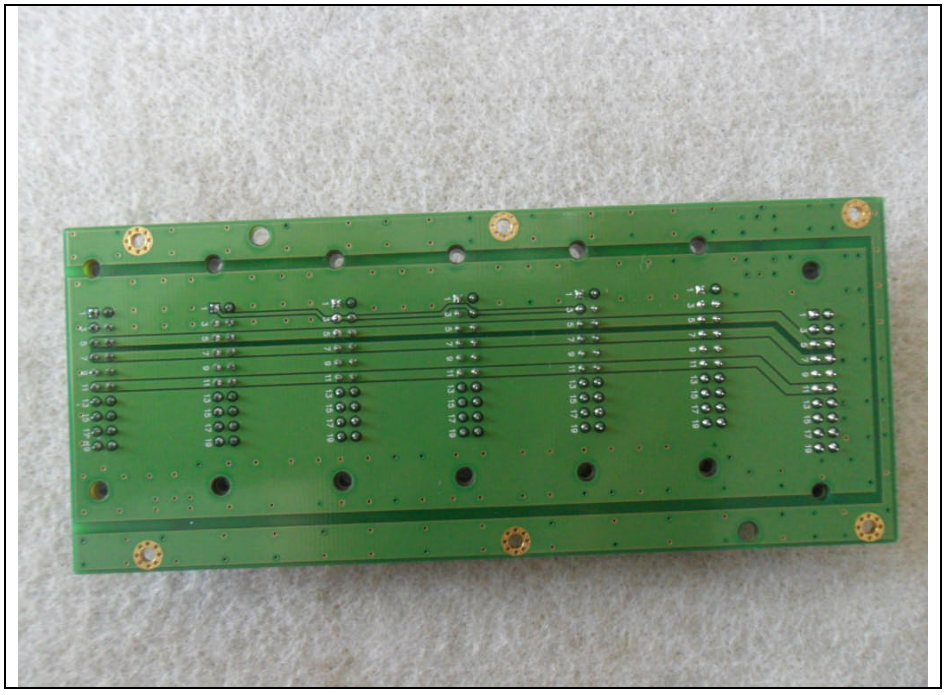
Inside



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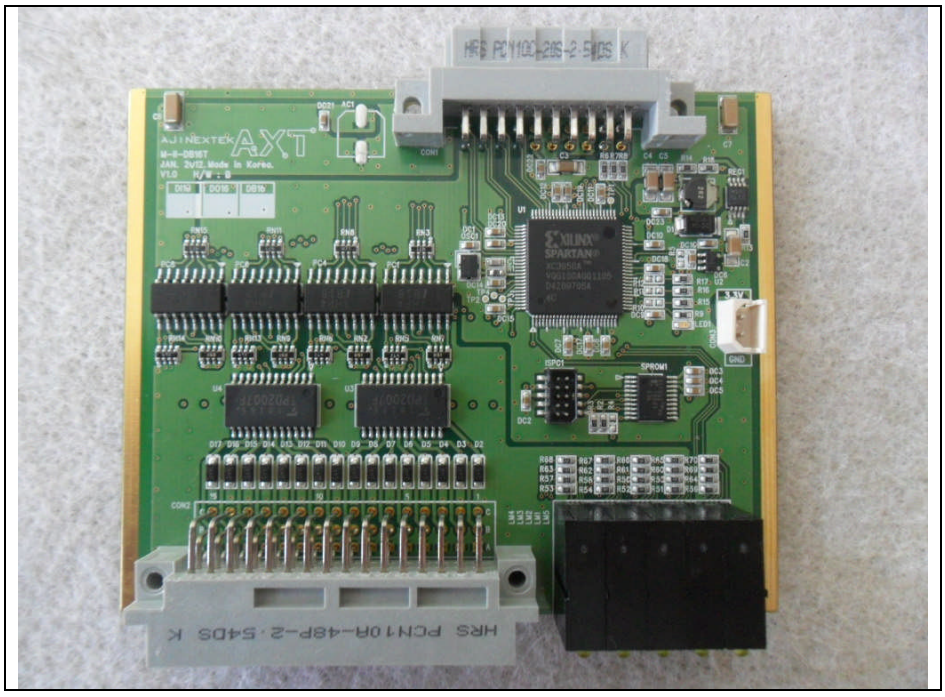
BACK PLANE

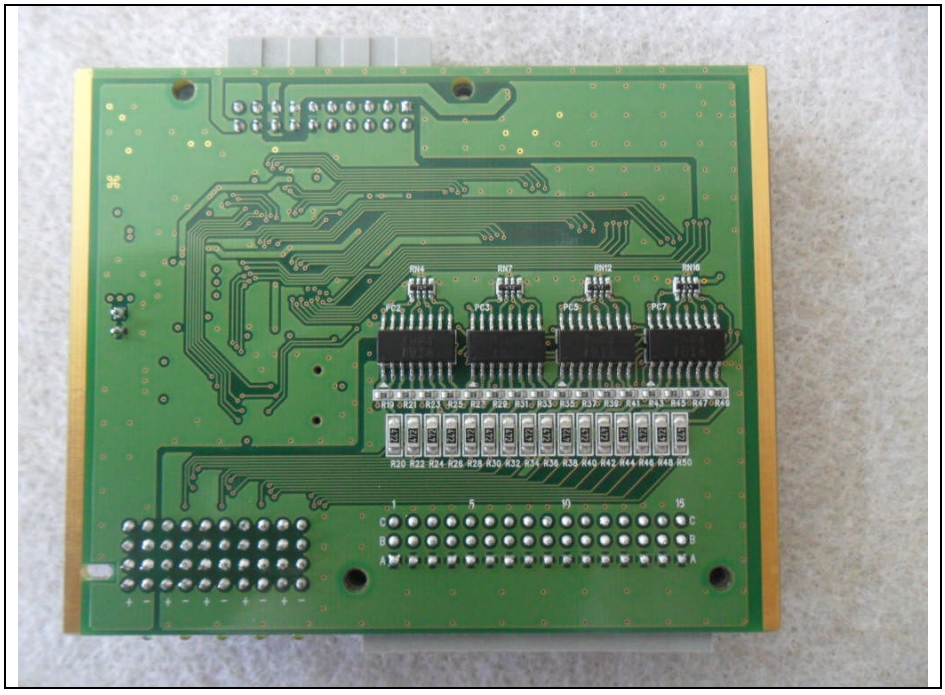




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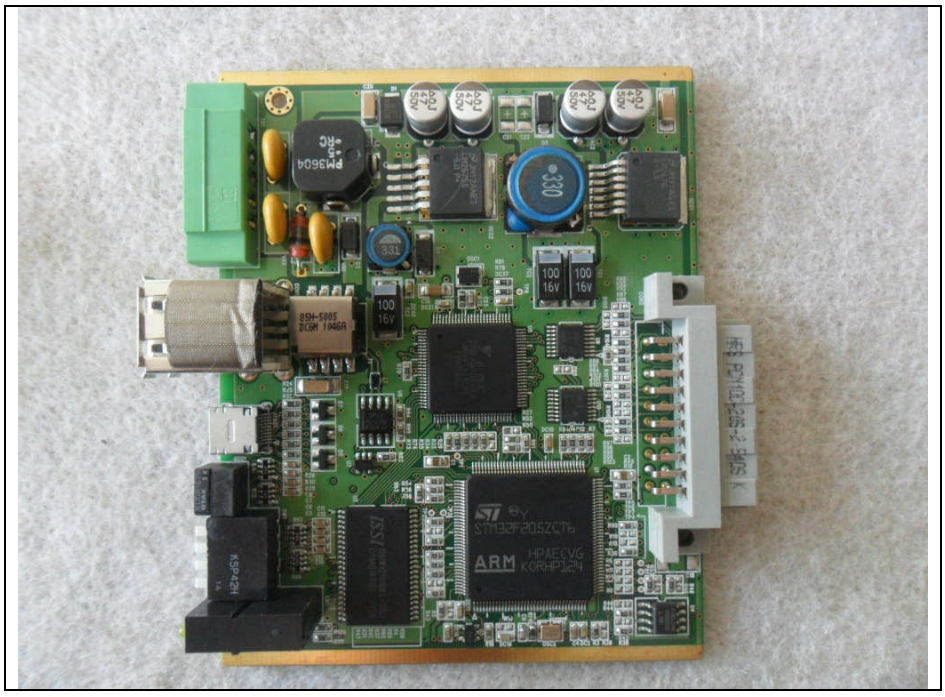
DB16T BOARD(6EA)

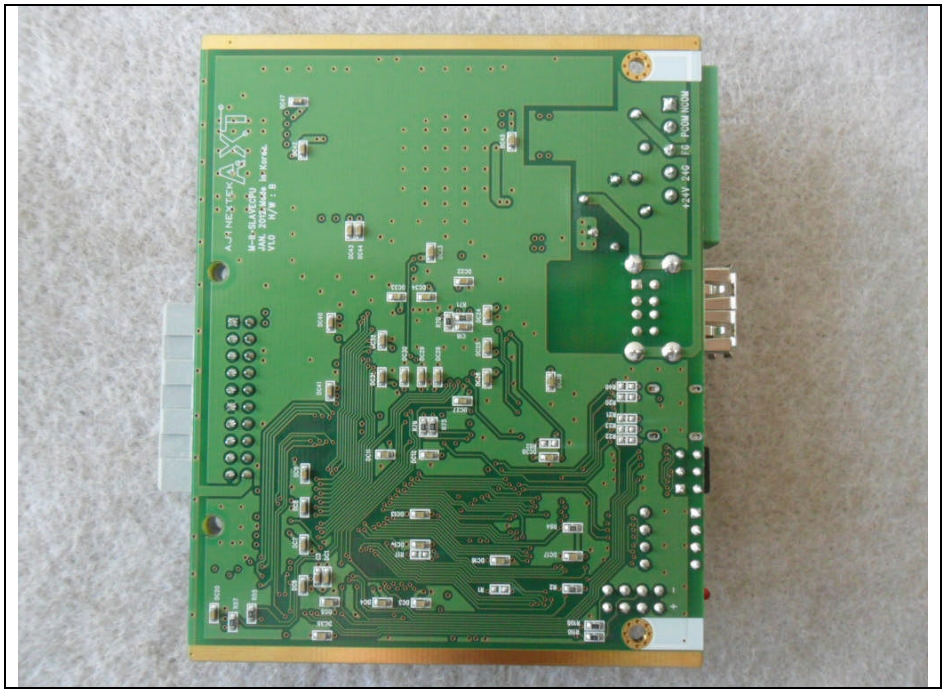




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SLAVE CPU BOARD

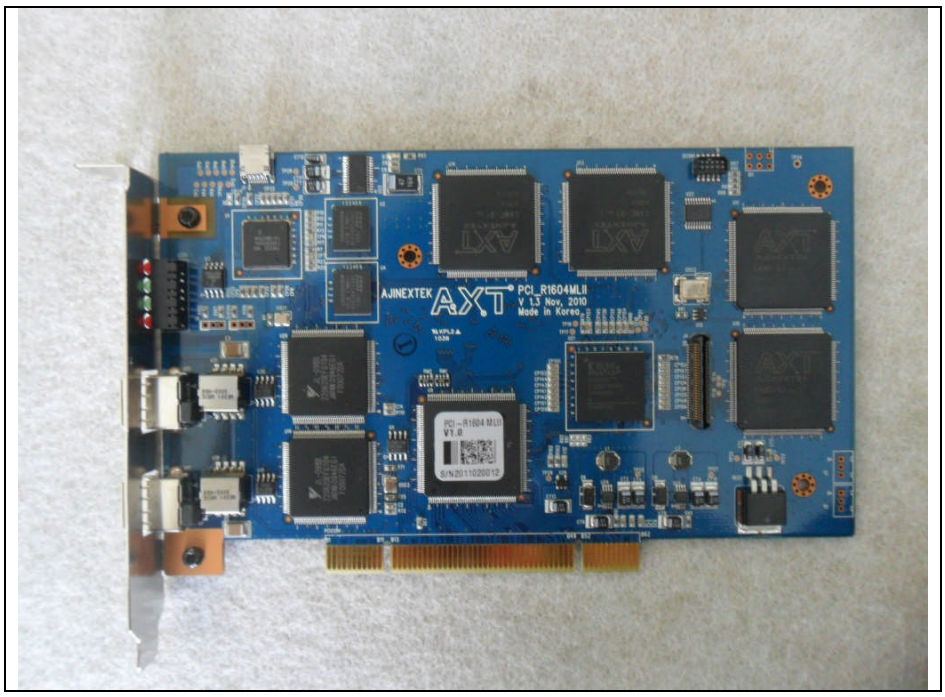




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* PCI-R1604 ML-II

Front View

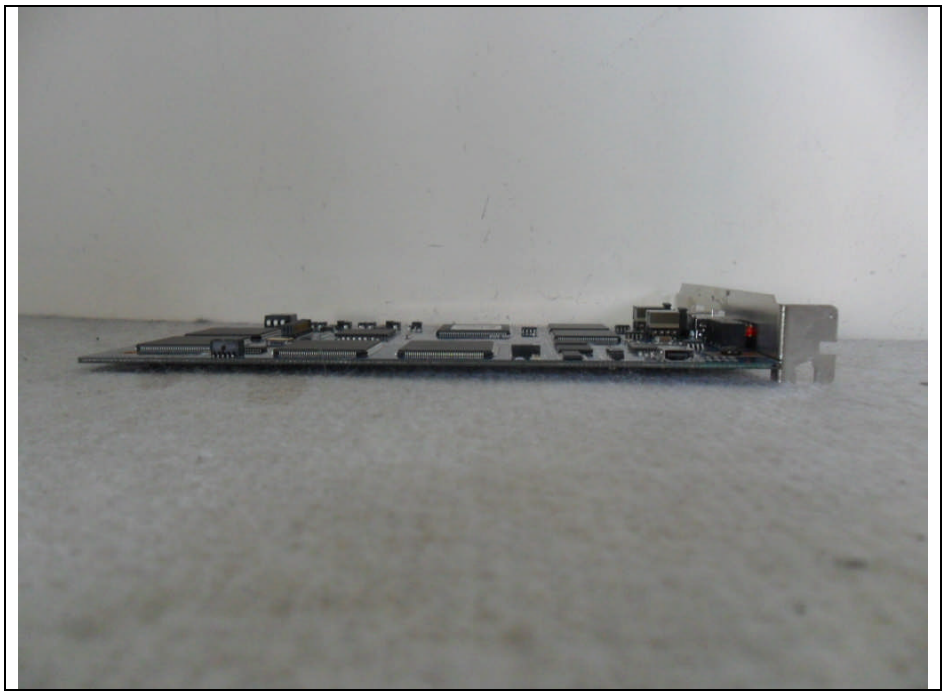


Rear View

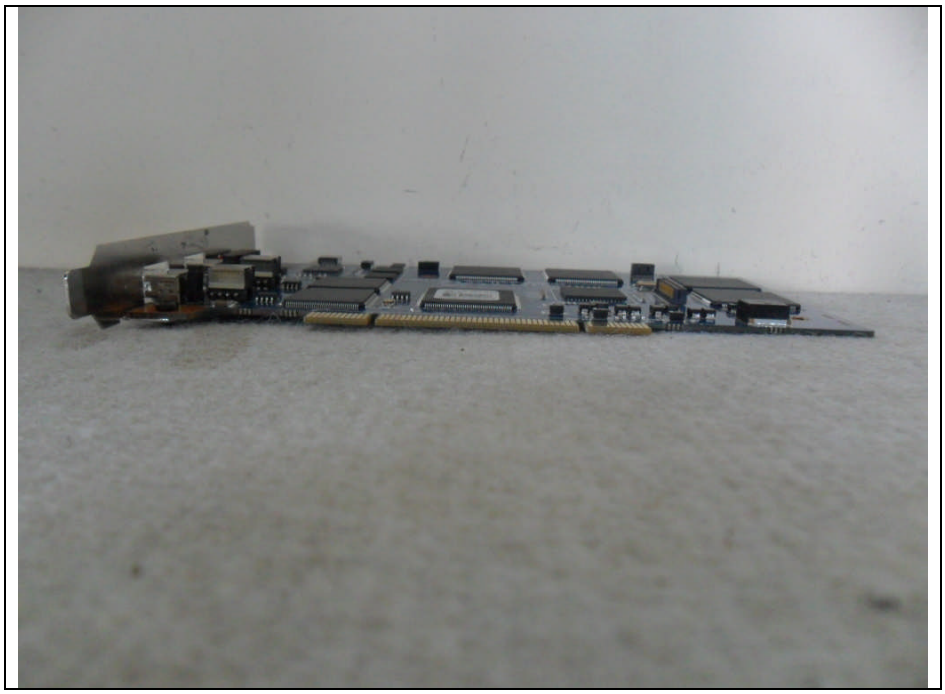


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Left View



Right View

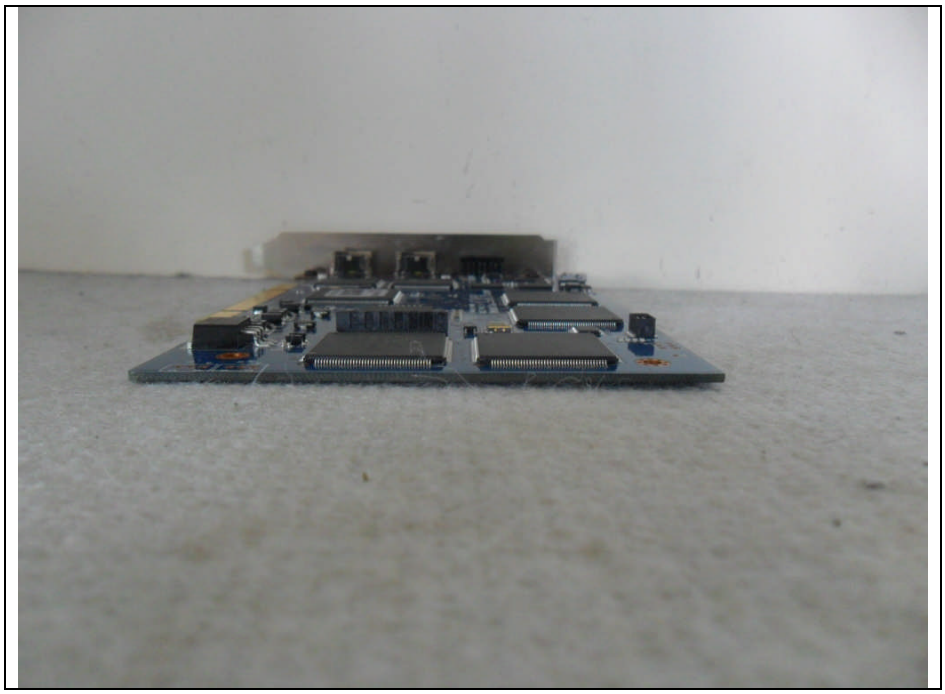


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Top View



Bottom View



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Inside (PC)



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USB Cable



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